USN | | | | | | | | | | [23CSE107]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

INTRODUCTION TO COMPUTER ARCHITECTURE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Examine the Boolean expression for the AND operation with two inputs, A and B	L2	CO1	**02**
	b.	Define bus in computer architecture	L1	CO2	**02**
	c.	What is addressing mode?	L1	CO3	**02**
	d.	What are the modes of data transfer?	L1	CO4	**02**
	e.	What is a write policy in cache memory?	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the basic operational concept of the CPU with an example	L2	CO1	**05**
		OR			
	b.	Explain the function of demultiplexer and its applications.	L2	CO1	**05**
3.	**a.**	Discuss the importance of Error Detection codes.	L2	CO2	**05**
		OR			
	b.	What are shift Micro operations? Discuss their types.	L2	CO2	**05**
4.	**a.**	Discuss the Floating-point and arithmetic representation.	L2	CO3	**05**
		OR			
	b.	Explain the role of peripheral devices in a computer system.	L2	CO3	**05**
5.	**a.**	Explain Direct Memory access (DMA) with a diagram.	L2	CO4	**05**
		OR			
	b.	Illustrate the concept of parallel processing and its applications	L2	CO4	**05**
6.	**a.**	Interpret the relationship between cache size and block size.	L2	CO5	**05**
		OR			
	b.	Explain the concept of memory hierarchy.	L2	CO5	**05**
7.	**a.**	Explain the importance of minimizing logic expressions with an	L2	CO1	**05**

Learning Resource Centre Jain Global Campus

		example.			
		OR			
	b.	Write a short note on PLA.	L2	CO1	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Formulate the structure and working of a flip-flop and discuss its various types.	L2	CO1	**10**
		OR			
	b.	Discuss the various functional units of a computer and their interactions during the program execution.	L2	CO1	**10**
9.	**a.**	Elaborate the Error detection codes and their working mechanisms.	L2	CO2	**10**
		OR			
	b.	Explain the numbers systems and give some examples for conversion.	L2	CO2	**10**
10.	**a.**	Describe Register Transfer Language in detail.	L2	CO3	**10**
		OR			
	b.	Explain different addressing modes with examples.	L2	CO3	**10**
11.	**a.**	Discuss asynchronous data transfer in detail, including handshaking and strobe control methods.	L2	CO4	**10**
		OR			
	b.	Discuss the priority interrupt mechanisms and its applications.	L2	CO4	**10**
12.	**a.**	What is memory management hardware? Explain its components.	L2	CO5	**10**
		OR			
	b.	Discuss cache memory mapping functions with examples.	L2	CO5	**10**
13.	**a.**	Explain Register transfer bus and memory transfers.	L2	CO3	**10**
		OR			
	b.	Explain the binary multiplication algorithm with an example.	L3	CO3	**10**